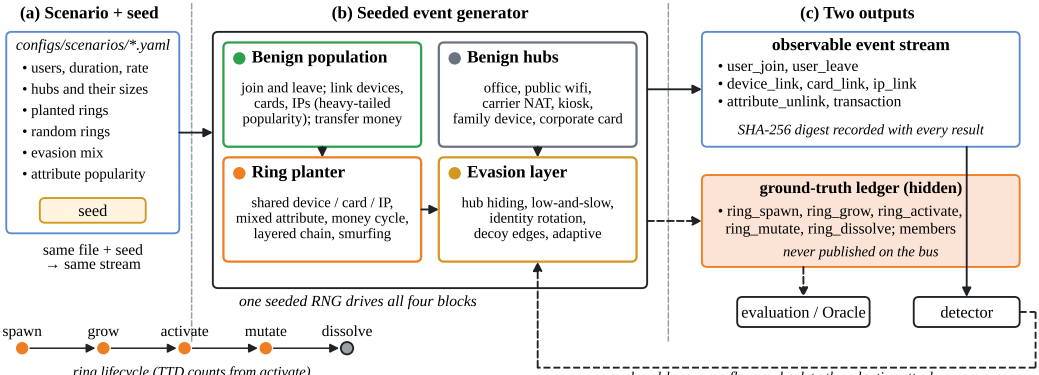
(a) Scenario + seed
configs/scenarios/*.yaml
• users, duration, rate
• hubs and their sizes
• planted rings
• random rings
• evasion mix
• attribute popularity
seed
same file + seed
→ same stream
(b) Seeded event generator
Benign population
join and leave; link devices, cards, IPs (heavy-tailed popularity); transfer money
Benign hubs
office, public wifi, carrier NAT, kiosk, family device, corporate card
Ring planter
shared device / card / IP, mixed attribute, money cycle, layered chain, smurfing
Evasion layer
hub hiding, low-and-slow, identity rotation, decoy edges, adaptive
one seeded RNG drives all four blocks
(c) Two outputs
observable event stream
• user_join, user_leave
• device_link, card_link, ip_link
• attribute_unlink, transaction
SHA-256 digest recorded with every result
ground-truth ledger (hidden)
• ring_spawn, ring_grow, ring_activate, ring_mutate, ring_dissolve; members
never published on the bus
evaluation / Oracle
detector
spawn
grow
activate
mutate
dissolve
ring lifecycle (TTD counts from activate)
closed-loop runs: flags go back to the adaptive attacker